Draft

MEK5960

Simon Lederhilger

February 10, 2026

Contents

Symbols **3**

1 Introduction **5**
1.1 Motivation . . . 5
1.2 Background . . . 5
1.3 Overview . . . 5

2 Vector fields **6**
2.1 Exterior calculus . . . 6
2.2 Complex analysis . . . 10
2.3 Fluid mechanics . . . 10
2.4 Potential fluid flow . . . 12
2.5 Vortical fluid flow . . . 13

3 Aerodynamics **15**
3.1 Boundary layers . . . 15
3.2 Airfoil theory . . . 15
3.3 Thin wing approximation . . . 16

4 Panel Methods **16**
4.1 The vortex lattice method . . . 17
4.2 Multipole expansions . . . 17
4.3 Octree data structure . . . 18
4.4 The fast multipole method . . . 18

5 Numerical Implementation **18**

6 Conclusion **18**

Symbols

Cyrillic symbols

Ƒ CHRISTOFFEL symbol 12

Greek symbols

$\Lambda\Omega$ exterior algebra on Ω, or sections thereof . 7

Ω pseudo-RIEMANN manifold 6

Latin symbols

$đ\mathfrak{a}$ exterior coderivative of $\mathfrak{a}$ 8

$T^*\Omega$ cotangent bundle of Ω, or sections thereof . 7

$d\mathfrak{a}$ exterior derivative of $\mathfrak{a}$ 8

$T\Omega$ tangent bundle of Ω, or sections thereof . . 7

Mathematical symbols

$\wedge$ exterior product 7

$\star$ HODGE star 8

The Cyrillic Alphabet

a	а	А	*o*	о	О
be	б	Б	*pe*	п	П
ve	в	В	*er*	р	Р
ge	г	Г	*es*	с	С
ǧe	ґ	Ґ	*te*	т	Т
de	д	Д	*će*	ћ	Ћ
đe	ђ	Ђ	*u*	у	У
e	е	Е	*ef*	ф	Ф
že	ж	Ж	*xa*	х	Х
ze	з	З	*ce*	ц	Ц
ež	ʒ	Ʒ	*če*	ч	Ч
i	и	И	*dže*	џ	Џ
je	й	Й	*ša*	ш	Ш
ka	к	К	*jery*	ы	Ы
el	л	Л	*jeri*	ь	Ь
lje	љ	Љ	*e*	э	Э
em	м	М	*ju*	ю	Ю
en	н	Н	*ja*	я	Я
nje	њ	Њ			

1 Introduction

1.1 Motivation

1.2 Background

1.3 Overview

We want to implement the fast multipole method, as presented in the article *Acceleration of Unsteady Vortex Lattice Method via Dipole Panel Fast Multipole Method* by Deng *et al.*.[1]

[1] [9] Deng *et al.* (2020)

2 Vector fields

Throughout this text, we shall draw from the results of the theory of differential geometry, the most important of which is *the fundamental theorem of multivariate calculus*. The conditions under which this theorem holds is of such importance to the validity of this work that some of the underlying theory will be discussed. Since a thorough understanding of the theory requires such attention to detail in order to discuss in any more meaningful detail than merely superficial mention, only the minimal required vocabulary and ideas to grasp will be introduced. For a more thorough derivation and explanation of the formalities of the theory and the background thereof, the reader is therefore refered to a dedicated text on the topic. The work at hand will utilize the conventions of Shigeyuki MORITA's *Geometry of Differential Forms* for this purpose, and any rigorous proofs or additional definitions to that end are referred thither. Admittedly, not including proofs of what is stated to be such foundational results preceding this work will not serve to enhance intuition *per se*, and so the reason for its inclusion is in part due to completeness, and in part so as to insure ourselves that there is indeed work from which we can draw already proven rules of arithmetic. Subsequently, the theory of functions of a complex variable will be discussed. For although the work at hand is concerned with fluid flow of three dimensions, results from the study of fluid flows of two dimensions will turn out to be useful, which we shall find can be viewed through the lens of complex analysis.

It ultimately is results pertaining to the fluid flow of air around and in the wake of a wind turbine rotor we wish to model. Of pertinence, then, is to outline the mathematical description of fluid flow altogether. For the purposes of this text, it is sufficient we establish for the moment a fluid to be a volume of a specified density and viscosity, both of which govern its dynamics, or motion. We suppose this motion may be represented in terms of a velocity vector of scalar functions in the infinite Euclidean space it occupies. In practice, such a vector is found through the integration of a differential equation governed by the applicable physical principles. We shall justify this notion of fluids for application to our problem of wind turbines, as it will in time become clear that the assumptions made are in no way general to the physics of fluid flow, or even correct, for that matter.

The aforementioned fundamental theorem of multivariate calculus may then be employed to transform such an integrated differential equation evaluated throughout the entire fluid domain to one over its boundary. In the case of a wind turbine blade, this entails integrating over its surface. There are two special cases of this fundamental theorem of particular interest, namely the GAUSS–OSTROGRADSKY theorem, and the THOMSON–STOKES theorem. The names are inconsequential, as they are indeed only merely offshoots of this more fundamental theorem, and are as such simply often labeled the divergence theorem and curl theorem. Unsurprisingly, they respectively relate the divergence and curl in the fluid to some flux of fluid at the boundary, and are given by

$$\int_\Omega \operatorname{div} \boldsymbol{u} \,\mathrm{d}V = \int_{\partial\Omega} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{S}, \qquad \int_\Omega \operatorname{curl} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{S} = \int_{\partial\Omega} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{s}. \tag{2.1a, b}$$

As is stated, the derivation of this fundamental theorem is beyond the scope of this work, so it will simply be presented once the foundation has been laid. This foundation will also reveal the theory of harmonic analysis, upon which we may build complex analysis. Once discussed, the topic of fluid mechanics will at last be addressed in full, and we shall discuss its dynamics and kinematics. Finally, we shall derive the theory with which this thesis seeks to address its problems—namely the decomposition of fluid processes in solenoidal and vortical porcesses.

2.1 Exterior calculus

The majority of the theoretical background in this work will hinge almost in its entirety on the validity of the equations (2.1), and theory sufficient for understanding their applications will be discussed in this section.

> **Definition 2.1** (Pseudo-RIEMANN manifold)**.** An n-dimensional manifold Ω is a topological space locally similar to Euclidean space $\mathbb{E}^n$, in a smooth sense. It is said to be of the pseudo-RIEMANN kind if it is equipped with the non-degenerate RIEMANN metric $\mathfrak{g}$.

The RIEMANN metric is left purposefully unspecified, as it depends on the representation of the manifold chosen. Throughout this work, we shall work the pesudo-RIEMANN manifold Ω of dimension 3, but we shall for the moment consider one of dimension n so as to speak of the details in general. The intrinsic smoothness of the manifold reveals something about its differentiability, which is defined by the tangent

space of some point in the manifold, $\mathrm{T}_p\,\Omega$. A tangent space $\mathrm{T}_p\,\Omega$ is technically a local space containing all tangent vectors at some point $p \in \Omega$,[1] so we shall introduce the *tangent bundle* as the disjoint union of such spaces for all points on the manifold, $\mathrm{T}\,\Omega = \sqcup_p\,\mathrm{T}_p\,\Omega$,[2] equipped with a natural projection $\Pi\colon\ \mathrm{T}\,\Omega \to \Omega$, such that vector fields $\boldsymbol{u} \in \mathrm{T}_p\,\Omega$ are mapped to $\Pi(\boldsymbol{u}) = p$.[3] Defining an inner product on this space, $\langle \boldsymbol{a}, \boldsymbol{b}\rangle = [\mathfrak{g}]_{ij} a^i b^j$, where EINSTEIN summation is implied, induces the metric.

In order to construct a meaningful notion of operations on the tangent bundle, we endow its dual, the cotangent bundle $\mathrm{T}^*\,\Omega$, the exterior product $\wedge$, forming the exterior algebra $\Lambda\Omega$. The exterior product may appear obtuse when first presented in the abstract manner we will shortly do, but rest assured it does indeed have a geometric interpretation. For, in an effort to unite the exterior algebra of Hermann GRASSMANN[4] and the quaternion algebra of William Rowan HAMILTON,[5] William Kingdon CLIFFORD laid the foundation on which the theory of the nominal CLIFFORD algebras were built.[6] On such algebras, the exterior product of k covectors is understood to be the oriented volume of the k-paralellotope bound by them,[7] and is just one part of the so-called *geometric product*, which provides a coherent definition of a vector product in general. We shall only draw from this intuition at a later point, as we shall focus on the properties of the exterior product with respect to objects on sections of the cotangent bundle.

Labeling the projection of a covector field $\mathfrak{u} \in \mathrm{T}^*\,\Omega$ in the same manner as a vector field, $\Pi^*(\mathfrak{u}) = p$, we define the section to be its inverse operation, uniquely returning the point to the covector field. The repeated application of the exterior product however admits higher grade covector fields. We will therefore use for a grade k monomial section of the exterior algebra $\mathfrak{a} \in \Lambda^k\Omega$ a k-form, and avoid the terminology of covectors altogether, as forms serve a much broader purpose to our end. The exterior product is alternating in the sense that for $\mathfrak{a}_i \in \Lambda\Omega$, given a permutation σ of the positive integers up to k, we have $\mathfrak{a}_{\sigma(1)} \wedge \cdots \wedge \mathfrak{a}_{\sigma(k)} = \operatorname{sgn}(\sigma)\left(\mathfrak{a}_1 \wedge \cdots \wedge \mathfrak{a}_k\right)$. The sign of a permutation $\operatorname{sgn}(\sigma)$ is 1 if the number of transpositions from the identity permutation is even, and -1 if the number of transpositions is odd. This alternating property further reduces to the two properties that

$$\mathfrak{a} \wedge \mathfrak{a} \equiv 0, \qquad \mathfrak{a} \wedge \mathfrak{b} \equiv -\mathfrak{b} \wedge \mathfrak{a}, \tag{2.2a, b}$$

for any forms $\mathfrak{a}, \mathfrak{b}$, of the same or different grade. To summarize, then:

> **Definition 2.2** (Exterior algebra)**.** Let $\mathrm{T}\,\Omega$ be a tangent bundle with basis ∂_i, and let $\wedge$ denote the product of the algebra $\Lambda\Omega = \mathrm{T}^*\,\Omega$. Then the members of the algebra
>
> $$\mathrm{d}x^i = \left([\mathfrak{g}]^{ij}\partial_j\right)^\flat$$
>
> form a basis for the cotangent bundle. When $\wedge : \Lambda^k\Omega \times \Lambda^l\Omega \to \Lambda^{k+l}\Omega$ satisfies the conditions of bilinearity and associativity, in addition to the modified alternating condition of equation (2.2b)
>
> $$\mathfrak{a} \wedge \mathfrak{b} = (-1)^{kl}\mathfrak{b} \wedge \mathfrak{a}, \tag{2.3}$$
>
> it is an exterior product, and $\Lambda^{k+l}\Omega$ is an exterior algebra.

Vector fields, then, as far as this work is concerned, are sections of the tangent space, though we will not notationally distinguish them. The notion of a tangent space is inherently connected to some sense of differentiability. Thus, given a differentiable curve on the manifold, the tangent vector at a point is described by this differentiability, and the partial derivative operators form a basis for the tangent space with respect to local coordinates on the manifold. We may choose for it some local basis $\{\boldsymbol{e}_j\}$, such that a vector field may be expressed $\boldsymbol{u} = u^j\boldsymbol{e}_j$. For $\mathbb{E}^3$, with coordinates x, y, z specifically, we shall always write its basis as $\{\partial_x, \partial_y, \partial_z\} \simeq \{\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}\}$. This *dual space* we discussed earlier may be regarded as the space of all linear functionals from the tangent space onto the field over which the vector field lies. Consider still the general basis vectors $\boldsymbol{e}_j$, along with some dual basis $\{\boldsymbol{e}^i\}$. As a result of the above discussion, we must have that $\boldsymbol{e}^i(\boldsymbol{e}_j) = \delta^i_j$. Clearly,

$$\boldsymbol{e}^i(\boldsymbol{u}) = \boldsymbol{e}^i(u^j\boldsymbol{e}_j) = u^j\boldsymbol{e}^i\boldsymbol{e}_j = u^j\delta^i_j = u^i.$$

In fact, any k-form is by construction a functional over the space of vector fields on the manifold. It is clear that there exists some isomorphism between the vectors and covectors. If $\mathfrak{u} \in \Lambda\Omega$ and $\boldsymbol{u} \in \mathrm{T}\,\Omega$

[1] [10] FLANDERS, p.54
[2] [20] JOST, p.12
[3] [27] MORITA, pp.169–171
[4] [12] GRASSMANN, §55
[5] [15] HAMILTON (1844)
[6] [8] CLIFFORD (1878)
[7] [12] GRASSMANN, pp.50–51

are isomorphic, these isomorphisms are usually denoted by the sharp $\mathfrak{u}^\sharp = \boldsymbol{u}$ and the flat $\boldsymbol{u}^\flat = \mathfrak{u}$, respectively raising and lowering the indices of the coefficients.[1] Stated explicitly, with these operators, we may describe any vector field $\boldsymbol{u} = u\hat{\boldsymbol{\imath}} + v\hat{\boldsymbol{\jmath}} + w\hat{\boldsymbol{k}}$ in terms of differential forms as $\boldsymbol{u}^\flat = u\,\mathrm{d}x + v\,\mathrm{d}y + w\,\mathrm{d}z$, so that $u^i\partial_i \simeq u_i\,\mathrm{d}x^i$. This notation however tends to clutter expressions and is cumbersome to read, and it will only be used when strictly necessary. As the astute reader may have noticed, we already readily distinguish between elements of the cotangent and tangent spaces by denoting the former with a fraktur typeface, and the latter with an italic bold. Thus it should be clear from context whichever of the two is meant at any given time.

We note the above expression implies the basis covectors, and indeed reiterate the fact that k-forms themselves are functions of vectors fields. In fact, when $\mathfrak{a} \in \Lambda\Omega$, we have that $\mathfrak{a}(\boldsymbol{u}) = a_i u^j$, which is a zero-form, or scalar. This generalizes to higher grade forms, with the introduction of the interior product, acting as a contraction in the usual sense of the term.

Definition 2.3 (Interior product)**.** Let $\boldsymbol{u} \in \mathrm{T}\,\Omega$, and $\mathfrak{a} \wedge \mathfrak{b} \in \Lambda^{k+l}\Omega$, noting the order of the addition of exponents in the exterior algebra. The interior product $\lrcorner : \mathrm{T}\,\Omega \times \Lambda^{k+l}\Omega \to \Lambda^{k+l-1}$, is given by

$$\boldsymbol{u} \lrcorner (\mathfrak{a} \wedge \mathfrak{b}) = (\boldsymbol{u} \lrcorner \mathfrak{a}) \wedge \mathfrak{b} + (-1)^k \mathfrak{a} \wedge \boldsymbol{u} \lrcorner \mathfrak{b}.$$

For unit $\mathfrak{b}$, we have $\mathfrak{a} \wedge 1 = \mathfrak{a}$, and so we say $\boldsymbol{u} \lrcorner \mathfrak{a} = \mathfrak{a}(\boldsymbol{u})$.

Extrapolating on this idea that the tangent space is inherently dependent on some notion of differentiability forming a basis for the tangent space, the cotangent space must then have a corresponding operation, which we will label the exterior derivative.

Definition 2.4 (Exterior derivative)**.** Given a manifold Ω and basis $\{\partial_i\}$, the exterior derivative $\mathrm{d} \colon \Lambda^k\Omega \to \Lambda^{k+1}\Omega$ of a k-form $\mathfrak{a} = a_{\dot{\imath}}\,\mathrm{d}x^{\dot{\imath}}$ is given by $\mathrm{d}\mathfrak{a} = \partial_i a_{\dot{\imath}}\,\mathrm{d}x^i \wedge \mathrm{d}x^{\dot{\imath}}$, where $\dot{\imath}$ is a multi-index of i. The product rule is graded, namely $\mathrm{d}(\mathfrak{a} \wedge \mathfrak{b}) = \mathrm{d}\mathfrak{a} \wedge \mathfrak{b} + (-1)^k \mathfrak{a} \wedge \mathrm{d}\mathfrak{b}$. Furthermore, $\mathrm{dd}\mathfrak{a} \equiv 0$.

In the manipulation of expressions of forms, it shall be necessary to convert forms of a higher grade to one of lower grade, and *vice versa*, for example before or after computing the exterior derivative. To that end, we introduce the Hodge star-operator.

Definition 2.5 (Hodge operator)**.** The Hodge operator $\star : \Lambda^k\Omega \to \Lambda^{n-k}\Omega$, where n is the dimension of Ω, such that $\mathfrak{a} \wedge \star\mathfrak{b} = \langle \mathfrak{a}, \mathfrak{b} \rangle\,\mathrm{d}V$. It holds that $\star\star\,\mathfrak{a} = (-1)^{k(n-k)}\mathfrak{a}$.

With the exterior derivative, we may describe the volume of the space with respect to local coordinates. In particular, for a manifold Ω of dimension n, $\mathrm{d}V \equiv \sqrt{|\mathfrak{g}|}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$, where $|\mathfrak{g}|$ is the determinant of the local metric. We find from the definition of the Hodge star above that $\star\mathrm{d}V = 1$.

In order to study the exterior derivative under the Hodge operator, we define the *codifferential*, given as follows.[2]

Definition 2.6 (Coderivative)**.** The codifferential of a k-form is adjoint to the exterior derivative, and is given by

$$đ\mathfrak{a} = (-1)^k \star^{-1} \mathrm{d} \star \mathfrak{a} = (-1)^{n(k+1)+1} \star \mathrm{d} \star \mathfrak{a}.$$

For illustrative purposes, the relationship between the exterior derivative, coderivative, and Hodge star is given by the following commutative diagram.

$$\begin{tikzcd}
\Lambda^k\Omega \arrow[r, "\star"] \arrow[d, "đ"'] & \Lambda^{n-k}\Omega \arrow[d, "\mathrm{d}"] \\
\Lambda^{k-1}\Omega \arrow[r, "(-1)^k\star"'] & \Lambda^{n-k+1}\Omega
\end{tikzcd}$$

[1] [28] Nishikawa, §3.1

[2] [27] Morita, §4.2

As we have noted, we are strictly speakning not performing these operations on the space $\Lambda^k\Omega$, but rather sections of it. On these sections we may define an inner product that should be familiar,

$$(\mathfrak{a}, \mathfrak{b}) = \int_\Omega \langle \mathfrak{a}, \mathfrak{b} \rangle \, \mathrm{d}V = \int_\Omega \mathfrak{a} \wedge \star\mathfrak{b}. \tag{2.4}$$

We have of course not defined what the integral of a differential form actually is. Still, assuming inheritance from our familiar notion of the integral as a linear operator, we find that it does indeed constitute an integral analogous to that of the L^2 inner product. This inner product is symmetric, and is the inner product with respect to which the codifferential is adjoint to the exterior derivative, in the sense that $(\mathrm{d}\mathfrak{a}, \mathfrak{b}) = (\mathfrak{a}, đ\mathfrak{b})$. For proofs and further discussion on the codifferential operator, the reader is again refered to Morita's monograph.[1]

We are now nearing the revelation of the fundamental theorem of multivariate calculus, and with it the end of this section. Before this, however, we ought to finally apply the operators to these differential forms to get a sense of the reason for which they were introduced to begin with. Consider the Euclidean vector field $\boldsymbol{u}^\flat = u\,\mathrm{d}x + v\,\mathrm{d}y + w\mathrm{d}z$. Upon calculating its exterior derivative $\mathrm{d}\mathfrak{u}$, one finds that

$$\mathrm{d}\mathfrak{u} = (\partial_x v - \partial_y u)\,\mathrm{d}x \wedge \mathrm{d}y + (\partial_z u - \partial_x w)\,\mathrm{d}z \wedge \mathrm{d}x + (\partial_y w - \partial_z v)\,\mathrm{d}y \wedge \mathrm{d}z.$$

This is nothing less than the two-form corresponding to the curl. Furthermore, the coderivative of a vector field is given by $đ\mathfrak{u} = -\star\mathrm{d}\star\mathfrak{u}$, which upon calculation one finds to be

$$đ\mathfrak{u} = -(\partial_x u + \partial_y v + \partial_z w).$$

This is of course the divergence with an opposite sign. And so, we define the operators to be such.

Definition 2.7 (Curl and divergence)**.** Let $\mathfrak{u} \in \Lambda^k\Omega$, with $\dim\Omega = 3$. The curl and divergence of $\mathfrak{u}$ are respectively given by

$$\operatorname{curl}\mathfrak{u} = \star\mathrm{d}\mathfrak{u}, \qquad \operatorname{div}\mathfrak{u} = \star\mathrm{d}\star\mathfrak{u}.$$

From these definitions, it is trivial to prove identities of great utility. For example, $\operatorname{div}\operatorname{curl}\mathfrak{u}$ and $\operatorname{curl}\operatorname{grad}\mathfrak{u}$ both being identically equal to zero. Another way to express the divergence is to consider the contraction of the gradient. We understand the trace of a matrix to be a form of contraction,[2] and the divergence of a vector field is simply the trace of the gradient.[3] For brevity, we take an *ad hoc* approach. Consider the exterior derivative of the interior product $\boldsymbol{u} \lrcorner\, \mathrm{d}V$. We calculate that

$$\begin{aligned} \mathrm{d}(\boldsymbol{u} \lrcorner\, \mathrm{d}V) &= \mathrm{d}\sum_i (-1)^{i+1} u_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n \\ &= \partial_i u_i \mathrm{d}V \\ &= \operatorname{div}\boldsymbol{u}\,\mathrm{d}V, \end{aligned} \tag{2.5}$$

where $\widehat{\mathrm{d}x^i}$ in the first equality in equation (2.5) means the term is omitted.

Having an idea of what the divergence and curl actually are, we are now ready for the progenitor of their named theorems—the fundamental theorem of multivariate calculus. The proof for this theorem is beyond the scope of the present work, so it will only be stated.

Theorem 2.8 (Fundamental Theorem of Multivariate Calculus)**.** *Let* $\mathfrak{u} \in \Lambda^{n-1}\Omega$*, and* $\partial\Omega$ *be the sufficiently regular boundary of* Ω*. Then,*

$$\int_{\partial\Omega} \mathfrak{u} = \int_\Omega \mathrm{d}\mathfrak{u}. \tag{2.6}$$

The convergence of integrals of the form (2.6) only makes sense for an unbounded fluid domain should the differential form $\mathfrak{u}$ evanesce sufficiently fast. This is clear from the condition that $\mathfrak{u}$ shall have compact support, meaning it would need to be a distribution. Such liberties are not ones we may always take, so we will make a concession in our definition of the manifold Ω. Suppose $\partial\Omega = S \cup \Sigma$, such that Σ encloses Ω, whilst S excises it, as is shown in figure 1. We may now extend Σ in such a way that Ω remains compact, and the general properties of $\mathfrak{u}$ are conserved, now only requiring sufficiently rapid evanescence.

[1] [27] Morita, pp.153–155

[2] expand on this.

[3] confirm this.

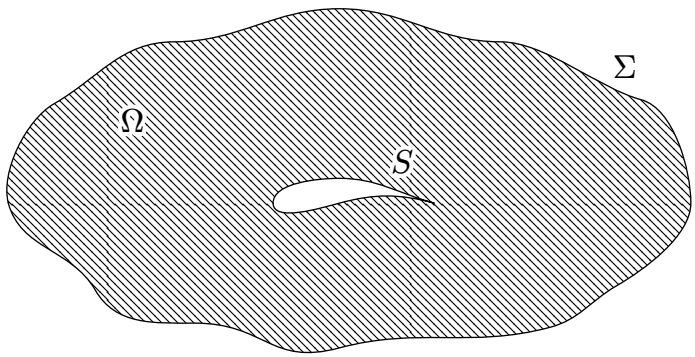

Figure 1: A cross section of Ω, illustrating the boundaries $S \cup \Sigma = \partial\Omega$.

An interesting inquiry to propose is of course how a punctured manifold behaves differently than one which is not. The answer is simple—problems will arise concerning the behavior of closed differential forms. We consider the following.

Lemma 2.9 (Poincaré)**.** *Let* $\mathfrak{a} \in \Lambda^k\mathbb{R}^n$ *be an arbitrary closed form, meaning* $\mathrm{d}\mathfrak{a} = 0$*. There then exists a form* $\mathfrak{b} \in \Lambda^{k-1}\mathbb{R}^n$ *such that* $\mathfrak{a} = \mathrm{d}\mathfrak{b}$*.*

The terminology for a form being the exterior derivative of some other form is *exact*. The Poincaré lemma states the manifold of figure 1, we can always find some local representation such that any closed form is exact.

As was noted above, the Gauss–Ostrogradsky theorem of equation (2.1a) is a special case of equation (2.6). To show this, we recall the result of equation (2.5), and consider the following equality.

$$\int_\Omega \mathrm{d}(\boldsymbol{u} \lrcorner \mathrm{d}V) = \int_{\partial\Omega} \boldsymbol{u} \lrcorner \mathrm{d}V.$$

Indeed, $\boldsymbol{u} \cdot \hat{\boldsymbol{n}}\mathrm{d}S = u\,\mathrm{d}y \wedge \mathrm{d}z - v\,\mathrm{d}z \wedge \mathrm{d}x + w\,\mathrm{d}x \wedge \mathrm{d}y = \boldsymbol{u} \lrcorner \mathrm{d}V$. And with that, the major requisite of the boundary element method is proved—namely its progenitor. It will be shown that the boundary element method applied to solenoidal fluid flow of no vorticity.

Definition 2.10 (Hodge–Laplace operator)**.** Let $\mathfrak{u} \in \Lambda^k\Omega$. We define the Hodge Laplacian to be

$$\Delta\mathfrak{u} = đ\mathrm{d}\mathfrak{u} + \mathrm{d}đ\mathfrak{u}. \tag{2.7}$$

Should $\Delta\mathfrak{u} = 0$, then $\mathfrak{u}$ is said to be harmonic.

Furthermore, we introduce a theorem of as great importance as the fundamental theorem of multivariate calculus.

Theorem 2.11 (Hodge decomposition)**.** *Any differential form* $\mathfrak{u}$ *may be decomposed as* $\mathrm{d}\mathfrak{a} + đ\mathfrak{b} + \mathfrak{c}$*, where* $\Delta\mathfrak{c} = 0$*.*

$$\iota : \partial\Omega \hookrightarrow \Omega$$

2.2 Complex analysis

An unbounded fluid, whose flow is invariant in a plane, will then be perfectly described by considering only the directions in which the fluid does vary. Examples of such flow are the cross section of an infinitely long cylinder, or any plane bisection a finite axisymmetric shape aligned with the flow.

2.3 Fluid mechanics

In our effort to mathematically describe natural fluid flow, our work shal be inspired by the principles of nature. The discussion that follows will elucidate to which extent our modeling actually reflects real fluids. Studying the physics of fluids is in large part shared by the consideration of the forces applied—dynamics—and the resulting motion—kinematics. It is the fluid kinematics we shall study

first. Regarding the motion of the fluid, we should like to distinguish two frames of reference, namely the material and the spatial. The material frame of reference follows some specific fluid particle, whilst the spatial frame observes the fluid passing through some fixed point.[1] Consider some particle $\boldsymbol{x}_0 \in \Omega_0$, which is carried by some flow onto the point $\boldsymbol{x} \in \Omega$ at time t. Distinguishing Ω_0 is a formality—for our purposes it is the very same manifold as we otherwise consider, and the flow is an automorphism. This configuration describes nothing less than the dynamical system $\boldsymbol{x} = \boldsymbol{c}(\boldsymbol{x}_0, t)$, $\boldsymbol{x}_0 = \boldsymbol{c}(\boldsymbol{x}_0, t)$, a solution of which is guaranteed locally by the Picard–Lindelöf theorem.[2] Clearly the velocity of the fluid at such a fixed point x must depend on the motion of the fluid particle. The nature of his dependency is however not immediately apparent. To this end, we consider the integral curves of the fluid velocity vector field $\boldsymbol{u}$ charted by $\boldsymbol{x}$. Such an integral curve we may in general label $\boldsymbol{c}(t)$, and are defined by its derivative coinciding with the velocity field at each point. These are not necessarily constrained by an initial condition, but do correspond with the flow when they are. Although we shall not do so, it may be shown that flows actually constitute a diffeomorphism group $\mathscr{G}$, a group of differentiable flows whose inverses exist and are differentiable. Upon inspection, the group action reveals itself to be map composition. The flow from some initial time by some increment of time will obviously constitute the same chronology when flowing therefrom to some other increment of time, should the flow start from the very same commencing position and be carried to the sum of the two considered time increments. We conclude that tracing the same path of the flow backwards constitutes applying the inverse flow map, yielding the composition of the flow with the inverse flow, being the identity operator. We label the derivative of the flow with respect to time the particle velocity $\boldsymbol{u}_{\boldsymbol{c}}$, representing the material frame of reference. Fixing some $\boldsymbol{x}$, it is now clear that the velocity in the spatial frame of reference is given by the expression $\boldsymbol{u} = \boldsymbol{u}_{\boldsymbol{c}} \circ \boldsymbol{c}^{-1}(\boldsymbol{x})$. Should an explicit formula for this relationship be established, equations of motion may be derived from Newtonian principles and deliberation of dynamics, as we would possess the constitutive acceleration. We shall, however, explore another avenue through Hamiltonian mechanics. Considerable amount of such work can be done from very minimal physical considerations, as is shown in Vladimir Arnol'd' s works.[3,4] Although the results of these works are magnificent, the process by which the are derived are beyond the scope of this work, and indeed the writer at the present time. We shall therefore draw from other monographs as sources on connexions and symplectic forms to present these results. Thus we cannot motivate all steps in the derivations to come. The geometric formulation of hydrodynamics developed in the periphery of these named works of Arnol'd do indeed build upon the Hamiltonian formalism using the theory of groups mentioned above. They consider variations of fluid paths, and seek to minimize their effort by means of observing the extrema of the so-called *action*. Considering the kinetic energy $\mathscr{L}(t) = {}^1\!/_2(\partial_t \boldsymbol{c}, \partial_t \boldsymbol{c})$.

$$\delta J = \frac{\mathrm{d}}{\mathrm{d}\epsilon} \tag{2.8}$$

We get the following theorem.[5]

> **Theorem 2.12** (Darboux)**.** *Let $\mathfrak{h}$ be a closed, nondegenerate bivector on some manifold of even dimension. Then every point $\boldsymbol{x}$ has some neighborhood on which we can find a coordinate system $(x^1, \dots, p_1, \dots)$, with*
>
> $$\mathfrak{h} = \mathrm{d}p_i \wedge \mathrm{d}x^i. \tag{2.9}$$

The meaning of this theorem is that we can not only guarantee that $\mathfrak{h}$ is closed, but also that it is exact, since we can write $\mathrm{d}p_i \wedge \mathrm{d}x^i = \mathrm{d}\mathfrak{p}$. Consider the projection $\Pi : \mathrm{T}^*\,\Omega \to \Omega$ such that the pullback is defined by $\mathrm{d}\Pi = \Pi_* : \mathrm{T}\,\mathrm{T}^*\,\Omega \to \mathrm{T}\,\Omega$. Then some tangent vector $\boldsymbol{a} \in \mathrm{T}_{(\boldsymbol{x},\mathfrak{p})}\,\Omega$ is given as $\boldsymbol{a} = a^i \partial_{x^i} + b_j \partial_{p_j}$, and $\Pi_* \boldsymbol{a} = a^i \partial_{x^i} \in \mathrm{T}_x\,\Omega$.

> **Definition 2.13** (Symplectic manifold)**.** A symplectic structure on a differentiable manifold of even dimension is a closed, nondegenerate bivector. The pair is a symplectic manifold.

$$\boldsymbol{u} \lrcorner\, \mathfrak{h} = -\mathrm{d}\mathscr{H}.$$

The symplectic form is preserved along the flow.

[1] [31] Truesdell, §14, pp.29–35
[2] [3] Arnol'd, §7.3, p.50
[3] [2] Arnol'd (1966)
[4] [4] Arnol'd & Xesin, ch.X
[5] [20] Jost, p.79

$$\frac{\partial x^i}{\partial t} = \frac{\partial \mathcal{H}}{\partial p_i}, \qquad \frac{\partial p_i}{\partial t} = -\frac{\partial \mathcal{H}}{\partial x^i}$$

Definition 2.14 (Lie derivative)**.** Let $\boldsymbol{u}$ be a vector field on the manifold Ω, and $\boldsymbol{c}$ the flow. For an arbitrary $\mathfrak{a} \in \Lambda^k\Omega$, the Lie derivative of $\mathfrak{a}$ by $\boldsymbol{u}$ is

$$\pounds_{\boldsymbol{u}}\mathfrak{a} = \partial_t|_{t=0}\boldsymbol{c}^*\mathfrak{a} = \mathrm{d}(\boldsymbol{u} \lrcorner \mathfrak{a}) + \boldsymbol{u} \lrcorner \mathrm{d}\mathfrak{a}.$$

The vector field $\boldsymbol{u}$ is symplectic if $\pounds_{\boldsymbol{u}}\mathfrak{a} = 0$ for a symplectic form $\mathfrak{a}$. For a vector field $\boldsymbol{v}$, the Lie derivative is defined to be

$$\pounds_{\boldsymbol{u}}\boldsymbol{v} = [\boldsymbol{u}, \boldsymbol{v}].$$

Definition 2.15 (Levi-Civita connexion)**.** Let $\boldsymbol{u} \in \mathrm{T}\,\Omega$ and $\mathfrak{a}$ be a section of some vector bundle. A connexion ∇ is a bilinear form. Should it preserve the metric in the sense that $\nabla\mathfrak{g} = 0$, and be torsion free, then it is a Levi-Civita connexion. These are determined by the Christoffel symbols

$$\nabla_i\partial_j = \Gamma^k_{ij}\partial_k, \qquad \Gamma^k_{ij} = \mathrm{g}^{k\ell}/2(\partial_j\mathrm{g}_{i\ell} + \partial_i\mathrm{g}_{j\ell} - \partial_\ell\mathrm{g}_{ij}). \tag{2.10}$$

Theorem 2.16 (Fundamental thoerem of pseudo-Riemannian geometry)**.** *Every pseudo-Riemannian manifold has a unique Levi-Civita connexion.*

A curve is called geodesic if $\nabla_{\dot{\boldsymbol{c}}}\dot{\boldsymbol{c}} = 0$.

The Euler equation is a geodesic equation of the right-invariant L^2 metric on the diffeomorphism group, where the pressure is the constraining force orthogonal to the diffeomorphism group.[1,2]

$$\varrho\mathrm{D}_t\boldsymbol{u} = \nabla \cdot \mathfrak{P} + \varrho\mathbf{g}, \qquad \mathrm{D}_t\varrho = -\varrho\nabla \cdot \boldsymbol{u} \tag{2.11a, b}$$

$$\varrho\partial_t\mathfrak{u} + \varrho\pounds_{\boldsymbol{u}}\mathfrak{u} = -\mathrm{d}p + \mu\Delta\mathfrak{u} + \varrho\mathbf{g}, \qquad đ\mathfrak{u} = 0 \tag{2.12a, b}$$

Theorem 2.17 (Reynolds transport)**.** *Let $\boldsymbol{u}$ be the vector field generated by the diffeomorphism $\boldsymbol{c}$, and $\mathfrak{a} \in \Lambda^k\Omega$. Then,*

$$\partial_t|_{t=0}\int_{\boldsymbol{c}(\Omega)}\mathfrak{a} = \int_\Omega \pounds_{\boldsymbol{u}}\mathfrak{a}. \tag{2.13}$$

2.4 Potential fluid flow

The generation of lift is undoubtedly a viscous phenomenon, fully realized by the well known Navier–Stokes and continuity equations.

Upon assuming conservative body forces, the Navier–Stokes equation then reduces to the Euler and incompressibility equations[3]

$$\varrho\mathrm{D}_t\boldsymbol{u} = -\nabla p, \qquad \nabla \cdot \boldsymbol{u} = 0. \tag{2.14a, b}$$

A vector field satisfying (2.14b) is called solenoidal. From the Hodge decomposition theorem, we may express the fluid velocity field as the sum of

$$\boldsymbol{u}^\flat = \mathrm{d}\Phi + đ\mathfrak{v} \tag{2.15}$$

It is apparent that the curl of the Euler equation (2.14) is zero, such that the vector field describing the fluid flow is harmonic.

$$\Delta\Phi = 0, \qquad \boldsymbol{x} \in \Omega \tag{2.16}$$

This representation of the fluid is manageable and allows for powerful tools with which to investigate it. It is known that the effects of viscosity are negligible for flow at large scale should the fluid not interact with an object.

[1] [33] Xesin & Modin (2025), Remark 2.3

[2] [19] Izosimov & Xesin (2024)

[3] discuss Mach number, incompressibility is very relevant here.

Consider two harmonic scalar functions ϕ and φ. Since all zero-forms are co-closed,[1] the HODGE–LAPLACE operator acting on zero-forms is equivalent to the familiar definition $\Delta\phi = \operatorname{div}\operatorname{grad}\phi$ of vector calculus. Consider the expressions $đ\phi d\varphi$ and $đ\varphi d\phi$. We take first the former, and expand the codifferential into its constituent stars and differential. From the definition of the HODGE star, we know it to be impotent acting on a scalar field. Thus,

$$đ\phi d\varphi = \star\langle d\phi, d\varphi\rangle\, dV + \phi\Delta\varphi = \star\langle d\varphi, d\phi\rangle\, dV = đ\varphi d\phi.$$

$$\int_{\partial\Omega} (\phi\nabla\psi - \psi\nabla\phi)\cdot\hat{\boldsymbol{n}}\, dS = 0. \tag{2.17}$$

This result is often referred to as GREEN's second identity. Constructing now a GREEN function defined to be harmonic everywhere, with the exception at some point $\boldsymbol{ж} \in \mathrm{Q}$, we may recast equation (2.17) in terms of the CAUCHY principal value.[2] For the case of three dimensional flow, the GREEN function is given by

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{ж}) = r^{-1}, \qquad r = |\boldsymbol{x} - \boldsymbol{ж}|. \tag{2.18a, b}$$

We confirm that $\Delta\,\mathrm{G}(\boldsymbol{x};\boldsymbol{ж}) = \delta(\boldsymbol{x} - \boldsymbol{ж})$, where δ is the DIRAC delta function, meaning the GREEN function of equation (2.18a) is harmonic almost everywhere. We may insert this function for ψ in equation (2.17), with the appropriate adjustments to the right-hand side. By integrating around the singularity at $\boldsymbol{ж}$, we find that

$$\mathrm{pv}\int_{\partial\Omega} (\Phi\partial_{\hat{\boldsymbol{n}}}\,\mathrm{G} - \mathrm{G}\,\partial_{\hat{\boldsymbol{n}}}\Phi)\, dS = k\pi\Phi(\boldsymbol{ж}), \qquad k = \begin{cases} 2, & \boldsymbol{ж} \in \partial\Omega \\ 4, & \boldsymbol{ж} \in \Omega \\ 0, & \text{otherwise} \end{cases}. \tag{2.19}$$

For the directional derivative, we mean $\partial_{\hat{\boldsymbol{n}}} = \hat{\boldsymbol{n}}\cdot\nabla$.

2.5 Vortical fluid flow

In order to study the effects of vortices on the velocity field, we define the vorticity $\mathfrak{w} = \star d\mathfrak{u}$, with the relation $\mathfrak{w}^\sharp = \boldsymbol{w}$. We consider the vorticity equation, obtained by taking the curl of the NAVIER–STOKES equation (2.12a).

$$\varrho\partial_t\mathfrak{w} + \varrho\pounds_{\boldsymbol{u}}\mathfrak{w} = \mu\Delta\mathfrak{w}, \qquad đ\mathfrak{w} = 0. \tag{2.20a, b}$$

Confer with BATCHELOR,[3] MARSDEN & WEINSTEIN,[4] or MODIN *et al.*[5] The question at hand is whether the vorticity is useful in the representation of the fluid flow we are interested in modeling. For any structure interacting with a fluid inducing a perturbed velocity, this velocity ought to not propagate indefinitely. In other words, it should be evanescent. Such vector fields may be described by decomposition into constituent HELMHOLTZ components, wherein

$$\boldsymbol{u} = \nabla\Phi + \nabla\times\boldsymbol{\Psi}, \qquad \boldsymbol{\Psi} = \psi\nabla A. \tag{2.21a, b}$$

We understand from the above discussion that this is just a special case of the HODGE decomposition, where Φ is a sharped zero-form, and $\boldsymbol{\Psi}$ is a sharped one-form. Indeed, as the curl operator is simply $\star d$, any choice of bivector $\mathfrak{v}$ such that $\star\mathfrak{v} = \Psi_i\, dx^i$ will be approporiate, revealing the familiar HODGE decomposition of theorem 2.11. Such a choice of representation of the vector potential is called a gauge, and the gauge chosen in equation (2.21b) is that of GUMEROV & DURAISWAMI,[6] where we impose the condition that $\operatorname{div}\boldsymbol{\Psi} = 0$. In other words, the bivector $\mathfrak{v}$ is exact. It ought to be noted that the condition of evanescence is not one which applies to the HELMHOLTZ decomposition, but rather that of the convergence of integrals which will be discussed promptly.

Should the fluid happen to interact with a structure, it is only near the surface the effects of viscosity are apparent. A formulation, then, wherein the vorticity is nonzero only on the boundary of the object, we may be able to proceed with a potential flow formulation in the rest of the fluid domain. Nevertheless, the laplacian term in the NAVIER–STOKES equation may be expanded in terms of equation (2.7) so as to reveal that

$$\mu\Delta\boldsymbol{u} = \mu\nabla(\nabla\cdot\boldsymbol{u}) - \mu\nabla\times\boldsymbol{w}. \tag{2.22}$$

[1] The proof is trivial. $đ\phi = \star d\star\phi = d\phi\wedge dV$, since the volume form is closed. The grade of $đ\phi$ is then not congruent with the dimension of the manifold, and thus vanishes.

[2] explain

[3] [5] BATCHELOR, §5.2, pp.267–273

[4] [23] MARSDEN & WEINSTEIN (1983), eq.V

[5] [25] MODIN *et al.* (2010)

[6] [14] GUMEROV & DURAISWAMI (2006)

This immediately yields the biharmonic equation for the potential function Φ when the vorticity is zero, which has been discussed by GUMEROV and DURAISWAMI in terms of implementation through fast multipole methods.[1] For we know the divergence of the velocity field to be zero, so that the Laplacian is equal to the curl of the vorticity. The connexion is immediately apparent between that of viscous effects encoded in the Laplacian term, and the vorticity of the fluid. Thus our motivation is plainly laid out before us—we wish to imitate the viscous effects that the fluid exhibits through a potential formulation. Employing the above HELMHOLTZ decomposition of equation (2.21), we find that by taking the laplacian thereof and substituting in the value of the vorticity of equation (2.22),

$$-\nabla \times \boldsymbol{w} = \nabla \times \Delta \boldsymbol{\Psi}.$$

We are free to equate the two arguments of the curl so that the vorticity equals the negative laplacian of the vector field $\boldsymbol{\Psi}$, or in terms of differential forms, $\Delta \star \mathfrak{v} = -\mathfrak{w}$. This is a POISSON equation, which is solved by the convolution $\star\mathfrak{v} = -\mathfrak{w} * \mathrm{G}$,[2] where G is the familiar GREEN function, expressed in local coordinates as

$$\Psi_i(\boldsymbol{x}) = -\int_{\Omega(\boldsymbol{ж})} w_i \, \mathrm{G} \, \mathrm{d}V, \qquad \mathrm{G}(\boldsymbol{x}, \boldsymbol{ж}) = \frac{1}{4\pi|\boldsymbol{x} - \boldsymbol{ж}|}.$$

Since it is the velocity $\mathfrak{u} = \operatorname{div} \mathfrak{v}$ we wish to find, we may simply calculate it from the above expression by taking the curl. Note that integration is done with respect to the variable $\boldsymbol{ж}$, whilst the exterior derivative will be taken with respect to $\boldsymbol{x}$. To that end, we have $\partial_i \mathfrak{w}(\boldsymbol{ж})\mathrm{d}x^i = 0$, so that by the graded product rule, we only differentiate the GREEN function. We then find the velocity vector field in terms of the vorticity, such that

$$\mathfrak{u}(\boldsymbol{x}) = -\int_\Omega \star\mathrm{d}(\mathfrak{w}\,\mathrm{G}) = -\int_\Omega \star(\mathfrak{w} \wedge \mathrm{d}\,\mathrm{G}\,).$$

Confer with XESIN.[3] As was noted above, the condition of evanescence of the velocity is a result of this formulation.[4]

We note the relation between the HODGE star of the exterior product of two 1-forms and the cross product of two vectors; $\star(\mathfrak{a} \wedge \mathfrak{b}) = (\boldsymbol{a} \times \boldsymbol{b})$, up to musical isomorphism. Expressing this in terms of vectors, we have the generalized BIOT–SAVART equation,

$$\boldsymbol{u}(\boldsymbol{x}) = -\frac{1}{4\pi}\int_\Omega \frac{\boldsymbol{w} \times (\boldsymbol{x} - \boldsymbol{ж})}{|\boldsymbol{x} - \boldsymbol{ж}|^3}\,\mathrm{d}V. \tag{2.23}$$

[1] *ibid.*
[2] [30] TAYLOR, §3.4
[3] [32] XESIN (2012)
[4] cite

Introduce circulation, such that the vorticity is given along vortex filaments, $\mathfrak{w} = \Gamma \mathfrak{t}$. Gives compact support, etc. so integrals converge nicely. Provide definition of circulation.

$$\Gamma = \oint_C \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{s} = \int_S \boldsymbol{w} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S.$$

We can derive the circulation theorem.

$$\mathrm{D}_t \Gamma = 0.$$

We will derive how boundary layers may be approximated by vortex sheets.

The Biot–Savart formulation of the disturbance velocity field at any point by the filamtent C is then given by[1,2]

$$\boldsymbol{u}(\boldsymbol{ж}) = \frac{1}{4\pi} \int_C \Gamma(\boldsymbol{x}) \frac{\boldsymbol{x} - \boldsymbol{ж}}{r^3} \times \mathrm{d}\boldsymbol{s}. \tag{2.24}$$

3 Aerodynamics

3.1 Boundary layers

It has been shown that for at least moderate Reynolds numbers, the fluid flow over an airfoil in two dimensions only depends on the curvature as a first order correction.[3]

3.2 Airfoil theory

Any blade of a wind turbine is composed from root to tip of a smooth distribution of cross sections, much like the one illustrated in figure 2a. Referred to as airfoils, these cross sections are optimized for the generation of lift. As was alluded to in the section on complex analysis, these are wholly two dimensional objects, whose ability to generate lift can be quantified by the means of this theory. Their mathematical representation warrants discussion, as it lends itself to the impending numerical analysis by way of comparison to theoretically derived values. Consider then the complex variable $\mathfrak{z}' = x' + iz'$ whose values trace the circle centered at the origin of radius c. The Žukvoskij transform takes a complex value and maps it to the point[4]

$$\mathfrak{z} = f(\mathfrak{z}') = \mathfrak{z}' + \frac{c^2}{\mathfrak{z}'}. \tag{3.1}$$

Expressing the variables in terms of their polar representations $x' = r\cos\theta$ and $z' = r\sin\theta$ yields

$$x = \left(r + \frac{c^2}{r}\right)\cos\theta, \qquad z = \left(r - \frac{c^2}{r}\right)\sin\theta. \tag{3.2a, b}$$

This constitutes an ellipse, as upon squaring the equations (3.2a, b), and adding them, we find that

$$\frac{x^2}{a^2} + \frac{z^2}{b^2} = 1, \qquad a = r + \frac{c^2}{r}, \qquad b = \left| r - \frac{c^2}{r} \right|. \tag{3.3a, b, c}$$

The airfoils of von Kármán & Trefftz are generalizations of those of Žukovskij.

$$\frac{\mathfrak{z} - 2c}{\mathfrak{z} + 2c} = \frac{(\mathfrak{z}' - c)^2}{(\mathfrak{z}' + c)^2} \tag{3.4}$$

It can be shown[5] that this is equivalent to the expression

$$\frac{\mathfrak{z}}{kc} = \frac{(\mathfrak{z}' + c)^k + (\mathfrak{z}' - c)^k}{(\mathfrak{z}' + c)^k - (\mathfrak{z}' - c)^k}. \tag{3.5}$$

[1] [9] Deng *et al.* (2020), eq.6
[2] [29] Schlichting & Truckenbrodt, p.98, eq.2.161
[3] [21] Lederhilger (2024)
[4] [29] Schlichting & Truckenbrodt, p.404, eq.6.37
[5] [24] Milne-Thomson, pp.125–128

(a) Žukovskij airfoil (b) von Kármán–Trefftz airfoil

Figure 2: Žukovsky and von Kármán–Trefftz airfoils, whose parameters are $c = 1.4$ and $s = {}^1\!/_4(i - {}^3\!/_5)$. The $k = 1.94$

3.3 Thin wing approximation

We shall for the rest of this work concern ourselves with the airfoil skeleton theory of Birnbaum, where the aerodynamical calculations are performed on the mean camber line only. Consider an airfoil whose chord coincides with the x-axis, given as a curve in the xz-plane. The mean camber line is the curve that for any value of x lies exactly midway between the two z-values of the airfoil.

In most practical scenarios, one would not be given a Žukovskij transform with which to analytically determine the mean camber line. One would instead be given an airfoil in terms of discrete points along its countour. The crucial observation to be made is that these points serve as vertices for a polygon inscribed in the airfoil. If one were given a distribution of vertices such that the number along the suction side and pressure side were equal, one could consider the midpoint along the diagonal of corresponding vertices, numbered from leading edge to trailing edge, or *vice versa*, as proposed by Hess.[1] Connecting each such midpoint from leading edge to trailing edge would yield an acceptable approximation to the mean camber line. Should the number of provided vertices be different on the suction and pressure sides, however, the procedure through which to obtain the mean camber line is not as evident. We shall determine an approximation to the mean camber line through studying the medial axes of polygons. Further, we conjecture these to be the optimal approximations, for the error of the medial axis to the mean camber line cannot be larger than the error of the inscribed polygon relative to the true curve of the airfoil. The medial axis was conceived of by Blum in a selcouth passage of writing on biological and visual shape description.[2] Its definition is as follows.[3]

> **Definition 3.1** (Medial axis)**.** Given a simple polygon $\Omega^\dagger$, its medial axis is the set of points $\{\boldsymbol{x}_i\}$ internal to it, such that there are at least two points on its boundary that are equdistant from and are closest to $\{\boldsymbol{x}_i\}$.

The peculiar choice of notation of the polygon stems from the interior of the wing, whose cross section is the airfoil at hand, being the complement of the manifold inhabiting the fluid of interest. Forcing a simple polygon, we employ the Jordan curve theorem,[4]

yielding a consistent notion of interior. The algorithm to be implemented is that of Lee, terminating branches of a Voronoj diagram, whose set of edges is a superset of the medial axis.[5]

Voronoj (Вороной) diagram?

Figure 3: Airfoil

4 Boundary Element Method

The boundary element method is a method of calculating values related to the fluid velocity through the second kind Fredholm integral equation (2.19), whereby the boundary is discretized into quadrilateral panels. As has been established, the conditions under which the solution is unique is determined at the boundary, here being the Neumann boundary condition of impermeability,

$$\partial_{\hat{\boldsymbol{n}}}\Phi(\boldsymbol{x}) = \hat{\boldsymbol{n}} \cdot \boldsymbol{U}, \qquad \boldsymbol{x} \in \partial\Omega. \tag{4.1}$$

We can now solve for the potential function Φ of equation (2.16) by the means we see fit. The alternative construction is to consider distributions of simple and double sources over the boundary. The potential

[1] [17] Hess (1974), see fig.7a
[2] [6] Blum (1967)
[3] [22] Li (1982),
[4] [11] Goluzin, pp.8&36
[5] [22] Li (1982), Corollary 3.

at the point $\boldsymbol{x}$ due to simple or double sources are given respectively by

$$\Phi(\boldsymbol{ж}) = \int_S \frac{\sigma(\boldsymbol{x})}{r}\,\mathrm{d}S, \qquad \Phi(\boldsymbol{ж}) = \int_S \frac{\mu(\boldsymbol{x})\cos\alpha}{r^2}\,\mathrm{d}S. \tag{4.2a ,b}$$

This is valid for any square integrable vector field.[1,2]

Imposing the condition of impermeability of equation (4.1) on the simple pole distribution, we find an integral equation for the source distribution. It can be shown that equation (4.1) can be recast in terms of an integral equation for te simple source distribution,

$$-2\pi\sigma + \int_S \sigma\partial_{\hat{\boldsymbol{n}}} r^{-1}\,\mathrm{d}S = \hat{\boldsymbol{n}}\cdot\boldsymbol{U}, \qquad \boldsymbol{x}\in\partial\Omega.$$

Confer with Hess & Smith.[3] This distribution can now be inserted into equation (4.2a) to find the potential.

4.1 The vortex lattice method

The vortex lattice method exchanges the doublet source panels of the boundary element method for vortex filaments along their boundaries. Multiple authors[4,5] have cited Hess & Smith's monumental work *Calculation of Potential Flow about Arbitrary Bodies*[6] to the equivalence between a dipole panel and a vortex panel. Such a statement is not found there. Rather, rather a relationship was demonstrated by Hess[7] between the velocity induced by a distribution of dipoles and that induced by a certain vortex distribution.

Some form of interpolation is done for the circulation on each filament, for which we choose a constant value. Thus,

$$\boldsymbol{u}(\boldsymbol{ж}) \approx \frac{1}{4\pi}\sum_{i=1}^{4}\Gamma_i \int_{C_i} \frac{\boldsymbol{x}-\boldsymbol{ж}}{r^3}\times\mathrm{d}\boldsymbol{s}.$$

For each filament, we can calculate the induced velocity. In order to reduce clutter, we drop the index indicating filament.

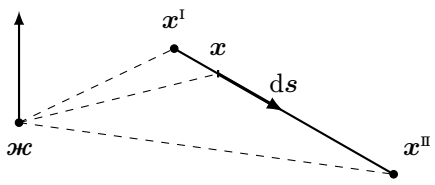

Figure 4: Line integral

We consider the parametrization of the line[8] such that $\mathrm{d}\boldsymbol{s} = {}^1\!/_2(\boldsymbol{x}^{\mathrm{I}} - \boldsymbol{x}^{\mathrm{II}})\,\mathrm{d}s$.

$$\boldsymbol{x} = \mathscr{x} + {}^s\!/_2(\boldsymbol{x}^{\mathrm{I}} - \boldsymbol{x}^{\mathrm{II}}), \qquad \mathscr{x} = \frac{x^{\mathrm{I}} + x^{\mathrm{II}}}{2} \tag{4.3a, b}$$

$$\boldsymbol{u}(\boldsymbol{ж}) = \frac{\Gamma}{4\pi}\frac{\boldsymbol{r}^{\mathrm{I}}\times\boldsymbol{r}^{\mathrm{II}}}{r^{\mathrm{I}}r^{\mathrm{II}} + \boldsymbol{r}^{\mathrm{I}}\cdot\boldsymbol{r}^{\mathrm{II}}}\left(\frac{1}{r^{\mathrm{I}}} + \frac{1}{r^{\mathrm{II}}}\right), \qquad \boldsymbol{r}_i = \boldsymbol{ж} - \boldsymbol{x}_i, \qquad \boldsymbol{x}_i \in \partial C_i. \tag{4.4}$$

4.2 Multipole expansions

We consider the points $\boldsymbol{x}$ and $\boldsymbol{ж}$ in $\mathbb{R}^3$, whose angle between them, of the triangle in the plane described by them and the origin, is denoted α. Although these points lie in three dimensional space, they form a unique plane, upon which the angle α may be defined. The points themselves may be given in terms of spherical coordinates, so that the distance between the points $\boldsymbol{x}$ and $\boldsymbol{ж}$ may be written in terms of the law of cosines,

$$r = |\boldsymbol{x} - \boldsymbol{ж}| = \sqrt{|\boldsymbol{x}|^2 - 2|\boldsymbol{x}||\boldsymbol{ж}|\cos\alpha + |\boldsymbol{ж}|^2}.$$

[1] [7] Byhovskiy & Smirnov (1960)
[2] [26] Morino (1993)
[3] [18] Hess & Smith (1967)
[4] [9] Deng *et al.* (2020)
[5] cite another
[6] [18] Hess & Smith (1967)
[7] [16] Hess (1972), Appendix A
[8] [13] Gumerov & Duraiswami (2013)

Given the above expression of the value r, we seek to approximate the GREEN function so that integrating it will be more efficient. It is clear from the above that we may write

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{ж}) = \frac{1}{|\boldsymbol{ж}|\sqrt{1+\varepsilon^2-2\varepsilon\cos\alpha}}, \quad \varepsilon = \frac{|\boldsymbol{x}|}{|\boldsymbol{ж}|}.$$

Through a binomial expansion in $\varepsilon^2 - 2\varepsilon\cos\alpha$, and rearranging so that we have a power series in ε, we have that

$$\mathrm{G}(\boldsymbol{x};\boldsymbol{ж}) = \frac{1}{|\boldsymbol{ж}|}\sum_{k=0}^{\infty}\mathrm{P}_k(\cos\alpha)\varepsilon^k, \quad \varepsilon < 1,$$

where P_k is the kth LEGENDRE polynomial.[1]

4.3 Octree data structure

An *octree* is a data structure which partitions the cube.

4.4 The fast multipole method

The fast multipole method calculates the velocity at some point $\boldsymbol{ж}$ using a sum of values using the methods of equation (?) for near field $\boldsymbol{x}$ and values using the multipole method. A local coordinate system is adopted, requiring translation and rotation of the origin, at which point a multipole expansion can be made. We suppose a dipole panel

$$\Phi(\boldsymbol{ж}) = -\frac{1}{4\pi}\int_S \frac{\boldsymbol{\mu}\cdot\boldsymbol{r}}{r^3}\,\mathrm{d}S,$$

where $\boldsymbol{\mu}$ is the dipole strength vector.

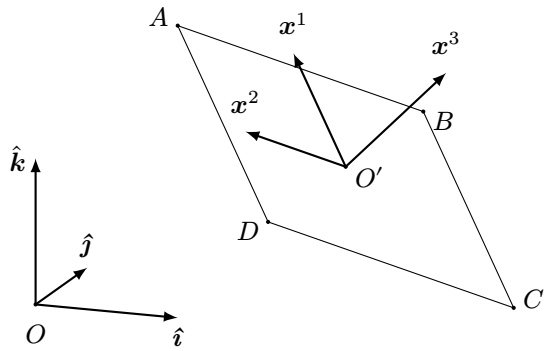

Figure 5: Panel coordinates

5 Numerical Implementation

6 Conclusion

References

[1] ABRAMOWITZ, Milton and STEGUN, Irene Ann. *Handbook of Mathematical Functions*. Third edition. National Bureau of Standards, 1965.

[2] ARNOL'D, Vladimir Igorevič (Арнольд). "Sur la géométrie différentielle des groups de LIE de dimension infinite et ses applications à l'hydrodynamique des fluides parfaits". In: *Annales de l'institut Fourier* 16.1 (1966), pp. 319–361.

[3] ARNOL'D, Vladimir Igorevič (Арнольд). *Ordinary Differential Equations (Обыкновенные дифференциальные уравнения)*. The MIT Press, 1973.

[1] [1] STEGUN, pp.332–353

[4] Arnol'd, Vladimir Igorevič (Арнольд) and Xesin, Boris Aronovič (Хесин). *Topological Methods in Hydrodynamics*. Second edition. Springer-Verlag, 2011.

[5] Batchelor, George Keith. *An Introduction to Fluid Dynamics*. Eigth edition. Cambridge University Press, 2007.

[6] Blum, Harry. "A Transformation for Extracting New Descriptions of Shape". In: *Models for the Perception of Speech and Visual Form*. Ed. by Weiant Wathen-Dunn. The MIT Press, 1967, pp. 362–380.

[7] Byxovskij, Èrik Borisovič (Быховский) and Smirnov, Nikolaj Vasil'evič (Смирнов). "б ортогональном разложении пространства векторфункций, квадратично суммируемых по заданной области, и операторах векторного анализа (On Orthogonal Expansions of the Space of Vector Functions Which are Square-summable over a Given Domain and the Vector Analysis Operators)". In: *Математические вопросы гидродинамики и магнитной гидродинамики для вязкой несжимаемой жидкости*. Vol. 59. Труды Математического института имени В. А. Стеклова. 1960, pp. 5–36.

[8] Clifford, William Kingdon. "Applications of Grassman's Extensive Algebra". In: *American Journal of Mathematics* 1.4 (1878), pp. 350–358.

[9] Deng, Shuai et al. "Acceleration of Unsteady Vortex Lattice Method via Dipole Panel Fast Multipole Method". In: *Chinese Journal of Aeronautics* 2.34 (2020), pp. 265–278.

[10] Flanders, Harley. *Differential Forms with Applications to the Physical Sciences*. Dover Publications, Inc., 1989.

[11] Goluzin, Gennadij Mixajlovič (Голузин). *Geometric Theory of Functions of a Complex Variable (Геометрическая теория функций комплексного переменного)*. The American Mathematical Society, 1968.

[12] Grassmann, Hermann. *Die lineale Ausdehnungslehre ein neuer Zweig der Mathematik*. Second. Verlag von Otto Wigand, 1878.

[13] Gumerov, Nail A. and Duraiswami, Ramani. "Efficient FMM Accelerated Vortex Methods in Three Dimensions via the Lamb–Helmholtz Decomposition". In: *Journal of Computational Physics* 240 (2013), pp. 310–328.

[14] Gumerov, Nail A. and Duraiswami, Ramani. "Fast Multipole Method for the Biharmonic Equation in Three Dimensions". In: *Journal of Computational Physics* 215 (2006), pp. 363–383.

[15] Hamilton, William Rowan. "On Quaternions. On a New System of Imaginaries in Algebra". In: *The London, Edinburgh and Dublin Philosophical Magazine and Journal of Science*. 3rd ser. XXV.169 (1844), pp. 489–495.

[16] Hess, John L. "Calculation of Potential Flow about Arbitrary Three-dimensional Lifting Bodies". In: *McDonnell Douglas Aircraft Company* MDC J5679-01 (1972).

[17] Hess, John L. "The Problem of Three-dimensional Lifting Potential Flow and its Solution by Means of Surface Singularity Distribution". In: *Computer Methods in Applied Mechanics and Enginering* 4 (1974), pp. 283–319.

[18] Hess, John L. and Smith, Apollo Milton Olin. "Calculation of Potential Flow about Arbitrary Bodies". In: *Progress in Aeronautical Sciences* 8 (1967). Ed. by Dietrich Küchemann, pp. 1–138.

[19] Izosimov, Anton Mixajlovič (Изосимов) and Xesin, Boris. "Geometry of Generalized Fluid Flows". In: *Calculus of Variations and Partial Differential Equations* 63.3 (2024).

[20] Jost, Jürgen. *Riemannian Geometry and Geometric Analysis*. Springer-Verlag, 2011.

[21] Lederhilger, Simon. *Boundary Layers on Airfoils*. Bachelor's thesis. 2024.

[22] Li, Decai (李德財). "Medial Axis Transformation of a Planar Shape". In: *Institute of Electrical and Electronics Engineers Transactions on Pattern Analysis and Machine Intelligence* PAMI-4.4 (1982).

[23] Marsden, Jerrold and Alan, Weinstein. "Coadjoint Orbits, Vortices, and Clebsch Variables for Incompressible Fluids". In: *Physica 7D* (1983), pp. 305–323.

[24] Milne-Thomson, Louis Melville. *Theoretical Aerodynamics*. Third edition. Macmillan & Co. Ltd., 1958.

[25] Modin, Klas et al. "Geodesics on Lie Groups: Euler Equations and Totally Geodesic Subgroups". In: *Research Letters in the Information and Mathematical Sciences* 14 (2010), pp. 79–106.

[26] Morino, Luigi. "Boundary Integral Equations in Aerodynamics". In: *Applied Mechanics Reviews* 46.8 (1993).

[27] Morita, Shigeyuki (森田茂之). *Geomtery of Differential Forms (微分形式の幾何学)*. The American Mathematical Society, 2001.

[28] Nishikawa, Seiki (西川青季). *Variational Problems in Geometry (幾何学的変分問題)*. The American Mathematical Society, 2002.

[29] Schlichting, Hermann and Truckenbrodt, Erich. *Aerodynamik des Flugzeuges, Erster Band*. Zweite Auflage. Springer-Verlag, 1967.

[30] Taylor, Michael Eugene. *Partial Differential Equations*. Third edition. Springer, 2023.

[31] Truesdell, Clifford. *The Kinematics of Vorticity*. Indiana University Press, 1954.

[32] Xesin, Boris. "Symplectic Structures and Dynamics on Vortex Membranes". In: *Moscow Mathematical Journal* 12.2 (2012), pp. 413–434.

[33] Xesin, Boris and Modin, Klas. *Universal Vector and Matrix Optimal Transport*. Oct. 2, 2025. arXiv: `2510.02039v1`.